

Contents

1	Basic	1
1.1	default	1
1.2	奇妙的東西	1
1.3	random	1
2	Math	1
2.1	basic	1
2.2	快速幂	1
2.3	大數運算	1
2.4	極座標排序	2
2.5	ExGCD	2
2.6	Phi	2
3	Binary Search	2
3.1	二分搜	2
3.2	三分搜	3
4	DataStructure	3
4.1	DSU(並查集)	3
4.2	FenwickTree/BIT	3
4.3	二維 BIT	3
4.4	Seg/線段樹	3
5	Graph	4
5.1	Topo/拓模	4
5.2	bellman/邊長鬆弛	4
5.3	ola/歐拉	4
5.4	LCA/最近公共祖先	4
6	dynamic programming	5
6.1	SOS DP	5
7	String	5
7.1	SATIS	5
8	酷東西可能會用到	5
8.1	areaofrect/矩形聯集面積	5
9	Math2	6
9.1	Theorem	6
9.2	Prime	6

1 Basic

1.1 default

```
#include <bits/stdc++.h>
using namespace std;
#define int long long
#define idonthevegirlfriend ios::sync_with_stdio(0),cin.tie(0);
#define pb push_back
#define endl "\n"
#define all(v) (v).begin(),(v).end()
void solve(){
}
signed main(){
    idonthevegirlfriend
    int _=1;
    //cin>>_;
    while(--){
        solve();
    }
}
```

1.2 奇妙的東西

```
priority_queue<T> //由大到小
priority_queue<T,vector<T>,greater<T>> //由小到大
__builtin_popcountll // 二進位有幾個1(記得這是long long)
__builtin_clzll // 左起第一個1之前0的個數
__builtin_parityll // 1的個數的奇偶性
__builtin_mul_overflow(a,b,&h) // a*b是否溢位,h = a * b
;
__builtin_add_overflow(a,b,&h)
```

1.3 random

```
mt19937 mt(chrono::steady_clock::now().time_since_epoch
().count());
//mt19937_64 mt() -> return randnum
int randint(int l, int r){
    uniform_int_distribution<> dis(l, r); return dis(mt
);
}
```

2 Math

2.1 basic

[a,b] 有包含
(a,b) 沒包含

2.2 快速幂

```
int ksm(int x, int y) {
    int ans = 1;
    while (y) {
        if (y & 1)
            ans = ans * x;
        x = x * x;
        y >>= 1;
    }
    return ans;
}

// 矩陣乘法
vector<vector<int>> mul(const vector<vector<int>> &A,
                      const vector<vector<int>> &B) {
    int n = A.size();
    vector<vector<int>> C(n, vector<int>(n, 0));

    for (int i = 0; i < n; i++) {
        for (int k = 0; k < n; k++) if (A[i][k]) {
            for (int j = 0; j < n; j++) {
                C[i][j] = (C[i][j] + A[i][k] * B[k][j])
                    % MOD;
            }
        }
    }
    return C;
}

// 矩陣快速幂
vector<vector<int>> ksm(vector<vector<int>> base, int
exp) {
    int n = base.size();
    vector<vector<int>> res(n, vector<int>(n, 0));

    for (int i = 0; i < n; i++) res[i][i] = 1;

    while (exp > 0) {
        if (exp & 1) res = mul(res, base);
        base = mul(base, base);
        exp >>= 1;
    }
    return res;
}
```

2.3 大數運算

```
//加法
string add(string num1, string num2) {
    string res = "";
    int i = num1.length() - 1, j = num2.length() - 1,
        carry = 0;
    while (i >= 0 || j >= 0 || carry) {
        int sum = carry + (i >= 0 ? num1[i--] - '0' :
0) + (j >= 0 ? num2[j--] - '0' : 0);
        carry = sum / 10;
        res += to_string(sum % 10);
    }
    reverse(res.begin(), res.end());
    return res;
}

//減法
string sub(string num1, string num2) {
    string res = "";
    int i = num1.length() - 1, j = num2.length() - 1,
        borrow = 0;
    while (i >= 0) {
        int sub = (num1[i--] - '0') - (j >= 0 ? num2[j--]
- '0' : 0) - borrow;
        if (sub < 0) {
            sub += 10;
            borrow = 1;
        } else {
            borrow = 0;
        }
    }
    return res;
}
```

```

        borrow = 0;
    }
    res += to_string(sub);
}
while (res.length() > 1 && res.back() == '0') res.
    pop_back(); // 去除前導0
reverse(res.begin(), res.end());
return res;
}

//乘法
string mul(string num1, string num2) {
    int n = num1.size(), m = num2.size();
    vector<int> res(n + m, 0);

    for (int i = n - 1; i >= 0; i--) {
        for (int j = m - 1; j >= 0; j--) {
            int p = (num1[i] - '0') * (num2[j] - '0');
            int sum = p + res[i + j + 1];

            res[i + j + 1] = sum % 10;
            res[i + j] += sum / 10;
        }
    }
    string ans = "";
    for (int x : res) {
        if (!ans.empty() && x == 0) // 跳過前導0
            ans += (char)(x + '0');
    }
    return ans.empty() ? "0" : ans;
}

//除法
bool smaller(string a, string b) {
    if (a.size() != b.size()) return a.size() < b.size();
    return a < b;
}

string div(string num1, string num2) {
    string cur = "", ans = "";
    for (char c : num1) {
        cur += c;
        // 去掉前導0
        while (cur.size() > 1 && cur[0] == '0') cur.
            erase(cur.begin());

        int x = 0;
        while (!smaller(cur, num2)) {
            cur = sub(cur, num2);
            x++;
        }
        ans += (char)(x + '0');
    }
    // 去掉前導0
    while (ans.size() > 1 && ans[0] == '0') ans.erase(
        ans.begin());
    return ans;
}

string mod(string num1, string num2) { //餘數
    string cur = "";

    for (char c : num1) {
        cur += c;
        while (cur.size() > 1 && cur[0] == '0') cur.
            erase(cur.begin());
        while (!smaller(cur, num2)) {
            cur = sub(cur, num2);
        }
    }
    return cur;
}

```

2.4 極座標排序

```

//由+x軸往逆時針排序
struct point {
    int x, y;
};
long long cross(const point &a, const point &b) {
    return (long long) a.x * b.y - (long long) a.y * b.
        x;
}

```

```

bool cmp(const point &a, const point &b) {
    int ah = (a.y < 0 or (a.y == 0 and a.x < 0));
    int bh = (b.y < 0 or (b.y == 0 and b.x < 0));
    if (ah != bh) return ah < bh;
    return cross(a, b) > 0;
}

void argument_sort(vector<point> &points) {
    sort(points.begin(), points.end(), cmp);
}

```

2.5 ExGCD

```

//ax+by=gcd(a,b)
PII gcd(int a, int b){
    if(b == 0) return {1, 0};
    PII q = gcd(b, a % b);
    return {q.second, q.first - q.second * (a / b)};
}

```

2.6 Phi

//歐拉函數 phi(n) 是指<=n中與n互質的數的個數

```

int _phi(int n) { // 0(sqrtN)
    int res = n, a = n;
    for (int i = 2; i * i <= a; i++) {
        if (a % i == 0) {
            res = res / i * (i - 1);
            while (a % i == 0) a /= i;
        }
    }
    if (a > 1) res = res / a * (a - 1);
    return res;
}

```

```

int phi[MXN]; // 建表 最大 1e7
void phi_table(int n) {
    phi[1] = 1;
    for (int i = 2; i <= n; ++i) {
        if (phi[i]) continue;
        for (int j = i; j <= n; j += i) {
            if (phi[j] == 0) phi[j] = j;
            phi[j] = phi[j] / i * (i - 1);
        }
    }
}

```

3 Binary Search

3.1 二分搜

```

int bsearch_1(int l, int r)
{
    while (l < r)
    {
        int mid = l + r >> 1;
        if (check(mid)) r = mid;
        else l = mid + 1;
    }
    return l;
}

// .....0000000000
int bsearch_2(int l, int r)
{
    while (l < r)
    {
        int mid = l + r + 1 >> 1;
        if (check(mid)) l = mid;
        else r = mid - 1;
    }
    return l;
}

// 000000000.....
int m = *ranges::partition_point(views::iota(0LL, (int)1
    e9+9), [&](int a){
    return check(a) > k;
});
//[begin,last)
//111111100000000000
//搜左邊數過來第一個 0
//都是 1 會回傳 last
int partitionpoint(int L,int R,function<bool(int)> chk)
{

```

```

int l = L, r = R - 1;
while(r - l > 10){
    int m = l + (r - l) / 2;
    if(chk(m)) l = m;
    else r = m;
}
int m = l;
while(m <= r){
    if(!chk(m)) break;
    ++m;
}
if(!chk(m)) return m;
else return R;
}
//手工

```

3.2 三分搜

```

int l = 1, r = 100;
while(l < r) {
    int lmid = l + (r - l) / 3; // l + 1/3區間大小
    int rmid = r - (r - l) / 3; // r - 1/3區間大小
    lans = cal(lmid), rans = cal(rmid);
    // 求凹函數極小值
    if(lans <= rans) r = rmid - 1;
    else l = lmid + 1;
}

```

4 DataStructure

4.1 DSU(並查集)

```

struct DSU {
    vector<int> f, sz;
    int cnt;
    DSU(int n) : f(n), sz(n) {
        for (int i = 0; i < n; i++) {
            f[i] = i;
            sz[i] = 1;
        }
    }
    int find(int x) {
        if (x == f[x]) return x;
        return f[x] = find(f[x]);
    }
    void merge(int x, int y) {
        x = find(x); y = find(y);
        if (x == y) return;
        if (sz[x] < sz[y])
            swap(x, y);
        f[y] = x;
        sz[x] += sz[y];
        cnt--;
    }
    bool same(int a, int b) {
        return find(a) == find(b);
    }
    int groups() { // 找有幾群
        return cnt;
    }
};

```

4.2 FenwickTree/BIT

```

struct Fenwick { // 這個是1base的
    int n;
    vector<int> s;
    int lowbit(int x) { return x & -x; }
    Fenwick(int _n) {
        n = _n + 1;
        s.clear(), s.resize(n, 0);
    }
    void update(int i, int v) { // 更新數值
        for (; i < n; i += lowbit(i))
            s[i] += v;
    }
    int query(int x) { // 查詢前綴和(有包含)
        int pre = 0;
        for (; x; x -= lowbit(x))
            pre += s[x];
        return pre;
    }
};

```

```

}
Fenwick(vector<int> a) { // 建構資料結構
    n = a.size(); // a 是 0-based
    s.clear(), s.resize(n + 1, 0); // s 是 1-based
    for (int i = 1; i <= n; i++)
        update(i, a[i - 1]);
}
// 使用:
Fenwick fw(a); // a is vector
Fenwick fw(n); // n is int

```

4.3 二維 BIT

```

struct BIT {
    int n;
    vector<vector<ll>> bit;
    int lowbit(int x) { return x & -x; }
    BIT(int _n) : n(_n + 1) {
        bit.assign(n, vector<ll>(n, 0));
    }
    void update(int x, int y, ll val) { // 更新 (x, y)
        for (int i = x; i < n; i += lowbit(i))
            for (int j = y; j < n; j += lowbit(j))
                bit[i][j] += val;
    }
    ll query(int x, int y) { // 查詢 (1, 1) ~ (x, y) 的和
        ll ans = 0;
        for (int i = x; i; i -= lowbit(i))
            for (int j = y; j; j -= lowbit(j)) ans += bit[i][j];
        return ans;
    }
    ll range_query(int a, int b, int x, int y) {
        return query(x, y) - query(x, b - 1) -
            query(a - 1, y) + query(a - 1, b - 1);
    }
};

```

4.4 Seg/線段樹

```

struct Seg {
#define cl(x) (x << 1)
#define cr(x) (x << 1) + 1
    int _n;
    vector<int> f, tag;
    vector<int> arr;
    // Seg(int n) : _n(n), f(4 * n), tag(4 * n) {}
    Seg(vector<int> a) {
        _n = a.size();
        f.resize(4 * _n + 1, 0);
        tag.resize(4 * _n + 1, 0);
        arr.resize(_n);
        for (int i = 0; i < _n; i++) {
            arr[i] = a[i];
        }
        build(1, 0, _n - 1);
    }
    Seg(int x) {
        _n = x;
        f.resize(4 * _n + 1, 0);
        tag.resize(4 * _n + 1, 0);
        arr.resize(_n);
    }
    void build(int id, int l, int r) {
        if (l == r) {
            f[id] = arr[l];
            return;
        }
        int mid = (l + r) >> 1;
        build(cl(id), l, mid);
        build(cr(id), mid + 1, r);
        pull(id, l, r);
    }
    void pull(int id, int l, int r) {
        int mid = (l + r) >> 1;
        push(cl(id), l, mid);
    }
};

```

```

    push(cr(id), mid + 1, r);
    f[id] = f[cl(id)] + f[cr(id)];
}

void push(int id, int l, int r) {
    if (tag[id]) {
        f[id] += tag[id] * (r - l + 1);
        if (l != r) {
            tag[cl(id)] += tag[id];
            tag[cr(id)] += tag[id];
        }
        tag[id] = 0;
    }
}

void update(int id, int l, int r, int x, int v) {
    if (l == r) {
        f[id] = v;
        return;
    }
    int mid = (l + r) >> 1;
    if (x <= mid) update(cl(id), l, mid, x, v);
    if (mid < x) update(cr(id), mid + 1, r, x, v);
    pull(id, l, r);
}

void update(int id, int l, int r, int ql, int qr,
            int v) {
    push(id, l, r);
    if (ql <= l and qr >= r) {
        tag[id] += v;
        return;
    }
    int mid = (l + r) >> 1;
    if (ql <= mid) {
        update(cl(id), l, mid, ql, qr, v);
    }
    if (qr > mid) {
        update(cr(id), mid + 1, r, ql, qr, v);
    }
    pull(id, l, r);
}

int query(int id, int l, int r, int sl, int sr) {
    push(id, l, r);
    if (sl <= l and sr >= r) {
        return f[id];
    }
    int res = 0;
    int mid = (l + r) >> 1;
    int ll = 0;
    int rr = 0;
    if (sl <= mid) {
        ll = query(cl(id), l, mid, sl, sr);
    }
    if (sr > mid) {
        rr = query(cr(id), mid + 1, r, sl, sr);
    }
    res = ll + rr;
    return res;
}
};

```

5 Graph

5.1 Topo/拓樸

```

vector<int> edge[MAXN], ans;
int deg[MAXN];

void topo(int n) {
    queue<int> q;

    for (int i = 1; i <= n; i++)
        if (!deg[i])
            q.push(i);

    while (!q.empty()) {
        int u = q.front();
        q.pop();
        ans.push_back(u);
    }
}

```

```

    for (int v : edge[u]) {
        deg[v]--;
        if (!deg[v])
            q.push(v);
    }
}
}

```

5.2 bellman/邊長鬆弛

```

struct Edge {
    int from, to, weight;
} edge[MAXN];
int dis[MAXN];
void bellman_ford(int n, int m) {
    dis[1] = 0; // 起點的距離一定是 0
    for (int j = 0; j < n - 1; j++) { // n 個節點要跑 n - 1 次
        for (int i = 0; i < m; i++) { // 對於所有邊都嘗試鬆弛
            if (dis[edge[i].to] >
                dis[edge[i].from] + edge[i].weight)
                dis[edge[i].to] =
                    dis[edge[i].from] + edge[i].weight;
        }
    }
}

```

5.3 ola/歐拉

```

//歐拉路徑(無向圖)兩個奇數點或者全偶點
//歐拉路徑(有向圖)條件:起點out=in+1,終點in=out+1,其他點
// in=out
//歐拉迴路(無向圖)所有點偶數點
//歐拉迴路(有向圖)所有點in=out
vector<int> anse; //邊
vector<int> ansp; //點
vector<bool> vis(m, false);
function<void(int)> dfs=&](int u){
    while(!mp[u].empty()){
        auto [x,e]=mp[u].back();
        mp[u].pop_back();
        if(vis[e]){
            continue;
        }
        vis[e]=true;
        dfs(x);
        anse.pb(e);
    }
    ansp.pb(u);
};

```

5.4 LCA/最近公共祖先

```

struct{
    int n, logN;
    vector<vector<int>>> jump;
    vector<int> in, out;
    bool is_ancestor(int x, int y){
        if(x == -1 || y == -1) return 1;
        return in[x] <= in[y] && out[y] <= out[x];
    }
    int query(int x, int y){
        if(is_ancestor(x, y)) return x;
        if(is_ancestor(y, x)) return y;
        for(int i = logN - 1; i >= 0; i--){
            if(!is_ancestor(jump[x][i], y)) x = jump[x][i];
        }
        return jump[x][0];
    }
}

void init(int _n, vector<vector<int>>>&v){ //size,
    adj array 0base
    n = _n;
    logN = log2(n) + 1;

    jump.resize(n, vector<int>(logN, -1));
    in.resize(n); out.resize(n);

    int timing = 1;
    auto dfs = [&](auto &&self, int x) -> void{

```

```

        in[x] = timing++;
        for(auto &i : v[x]){
            jump[i][0] = x;
            self(self, i);
        }
        out[x] = timing++;
        return;
    }; dfs(dfs, 0);

    for(int i = 1; i < logN; i++){
        for(int now = 1; now < n; now++){
            if(jump[now][i-1] == -1) continue;
            jump[now][i] = jump[jump[now][i-1]][i-1];
        }
    }
}
}LCA;

```

6 dynamic programming

6.1 SOS DP

```

// 求子集和 或超集和 -> !(mask & (1 << i))
for(int i = 0; i < (1<<N); ++i) F[i] = A[i]; //預處理 狀態權重

for(int i = 0; i < N; ++i)
    for (int s = 0; s < (1<<N); ++s)
        if (s & (1 << i))
            F[s] += F[s ^ (1 << i)];

//窮舉子集合
for(int s = mask; s; s = (s-1)&mask;)
//放這個做啥我不會用QAQ

```

7 String

7.1 SAIS

```

// 此為後綴陣列模板
// 用法:
// 把要處理的字串轉成整數陣列傳入suffix_array(), 會得到SA與H維結果
const int N = 500010; // 有可能要在這就寫原本字串的兩倍長
struct SA {
#define REP(i, n) for (int i = 0; i < int(n); i++)
#define REP1(i, a, b) for (int i = (a); i <= int(b); i++)
    bool _t[N * 2];
    int _s[N * 2], _sa[N * 2], _c[N * 2], x[N], _p[N],
        _q[N * 2], hei[N], r[N];
    int operator[](int i) { return _sa[i]; }
    void build(int* s, int n, int m) {
        memcpy(_s, s, sizeof(int) * n);
        sais(_s, _sa, _p, _q, _t, _c, n, m);
        mkhei(n);
    }
    void mkhei(int n) {
        REP(i, n) r[_sa[i]] = i;
        hei[0] = 0;
        REP(i, n) if (r[i]) {
            int ans = i > 0 ? max(hei[r[i] - 1] - 1, 0) : 0;
            while (_s[i + ans] == _s[_sa[r[i] - 1] + ans])
                ans++;
            hei[r[i]] = ans;
        }
    }
    void sais(int* s, int* sa, int* p, int* q, bool* t,
        int* c, int n, int z) {
        bool uniq = t[n - 1] = true, neq;
        int nn = 0, nmzx = -1, *nsa = sa + n, *ns = s + n,
            lst = -1;
#define MS0(x, n) memset((x), 0, n * sizeof(*(x)))
#define MAGIC(XD) \
        MS0(sa, n); \
        memcpy(x, c, sizeof(int) * z); \
        XD; \
        memcpy(x + 1, c, sizeof(int) * (z - 1)); \
        REP(i, n)

```

```

        if (sa[i] && !t[sa[i] - 1]) \
            sa[x[sa[i] - 1]++] = sa[i] - 1; \
        memcpy(x, c, sizeof(int) * z); \
        for (int i = n - 1; i >= 0; i--) \
            if (sa[i] && t[sa[i] - 1]) \
                sa[--x[sa[i] - 1]] = sa[i] - 1;
        MS0(c, z);
        REP(i, n) uniq &= ++c[s[i]] < 2;
        REP(i, z - 1) c[i + 1] += c[i];
        if (uniq) {
            REP(i, n) sa[--c[s[i]]] = i;
            return;
        }
        for (int i = n - 2; i >= 0; i--)
            t[i] =
                (s[i] == s[i + 1] ? t[i + 1] : s[i] < s[i + 1]);
        MAGIC(REP1(i, 1, n - 1) if (t[i] && !t[i - 1])
            sa[--x[s[i]]] = p[q[i] = nn++] = i);
        REP(i, n) if (sa[i] && t[sa[i]] && !t[sa[i] - 1]) {
            neq = lst < 0 || memcmp(s + sa[i], s + lst,
                (p[q[sa[i]] + 1] - sa[i]) *
                    sizeof(int));
            ns[q[lst = sa[i]]] = nmzx += neq;
        }
        sais(ns, nsa, p + nn, q + n, t + n, c + z, nn,
            nmzx + 1);
        MAGIC(for (int i = nn - 1; i >= 0; i--)
            sa[--x[s[p[nsa[i]]]]] = p[nsa[i]]);
    }
} sa;
int H[N], SA[N]; // SA: 後綴陣列的length
// H[i]: SA[i]與SA[i - 1]的共同子字串長
void suffix_array(int* ip, int len) {
    // should padding a zero in the back
    // ip is int array, len is array length
    // ip[0..n-1] != 0, and ip[len] = 0
    ip[len++] = 0;
    sa.build(ip, len, 128);
    for (int i = 0; i < len; i++) {
        H[i] = sa.hei[i + 1];
        SA[i] = sa._sa[i + 1];
    }
    // resulting height, sa array \in [0,len)
}

```

8 酷東西可能會用到

8.1 areaofrect/矩形聯集面積

```

struct AreaofRectangles{
#define cl(x) (x<<1)
#define cr(x) (x<<11)
    int n, id, sid;
    pair<int,int> tree[MXN<<3]; // count, area
    vector<int> ind;
    tuple<int,int,int,int> scan[MXN<<1];
    void point(int i, int l, int r){
        if(tree[i].first) tree[i].second = ind[r+1] - ind[l]; //當前區間有被做加值
        else if(l != r){ //沒有就直接拿兩個子節點的第二個數字
            int mid = (l+r)>>1;
            tree[i].second = tree[cl(i)].second + tree[cr(i)].second;
        }
        else tree[i].second = 0;
    }
    void upd(int i, int l, int r, int ql, int qr, int v){
        if(ql <= l && r <= qr){
            tree[i].first += v;
            point(i, l, r); return;
        }
        int mid = (l+r) >> 1;
        if(ql <= mid) upd(cl(i), l, mid, ql, qr, v);
        if(qr > mid) upd(cr(i), mid+1, r, ql, qr, v);
        point(i, l, r);
    }
}

```

```

void init(int _n){
    n = _n; id = sid = 0;
    ind.clear(); ind.resize(n<<1);
    fill(tree, tree+(n<<2), make_pair(0, 0));
}
void addRectangle(int lx, int ly, int rx, int ry){
    ind[id++] = lx; ind[id++] = rx;
    scan[sid++] = make_tuple(ly, 1, lx, rx);
    scan[sid++] = make_tuple(ry, -1, lx, rx);
}
int solve(){
    sort(ind.begin(), ind.end());
    ind.resize(unique(ind.begin(), ind.end()) - ind
        .begin()); // 離散化
    sort(scan, scan + sid);
    int area = 0, pre = get<0>(scan[0]);
    for(int i = 0; i < sid; i++){
        auto [x, v, l, r] = scan[i];
        area += tree[1].second * (x-pre); //
        upd(1, 0, ind.size()-1, lower_bound(ind.
            begin(), ind.end(), l)-ind.begin(),
            lower_bound(ind.begin(), ind.end(), r)-
            ind.begin()-1, v);
        pre = x;
    }
    return area;
}
} rect;

```

9 Math2

9.1 Theorem

- Lucas's Theorem :
For $n, m \in \mathbb{Z}^*$ and prime P , $C(m, n) \bmod P = \prod C(m_i, n_i)$ where m_i is the i -th digit of m in base P .
- Stirling approximation :
$$n! \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n e^{\frac{1}{12n}}$$
- Stirling Numbers(permutation $|P| = n$ with k cycles):
 $S(n, k) = \text{coefficient of } x^k \text{ in } \prod_{i=0}^{n-1} (x+i)$
- Stirling Numbers(Partition n elements into k non-empty set):
$$S(n, k) = \frac{1}{k!} \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n$$
- Pick's Theorem : $A = i + b/2 - 1$
 A : Area, i : grid number in the inner, b : grid number on the side
- Catalan number : $C_n = \binom{2n}{n} / (n+1)$
$$C_n^{n+m} - C_{n+1}^{n+m} = (m+n)! \frac{n-m+1}{n+1} \quad \text{for } n \geq m$$

$$C_n = \frac{1}{n+1} \binom{2n}{n} = \frac{(2n)!}{(n+1)!n!}$$

$$C_0 = 1 \quad \text{and} \quad C_{n+1} = 2 \binom{2n+1}{n+2} C_n$$

$$C_0 = 1 \quad \text{and} \quad C_{n+1} = \sum_{i=0}^n C_i C_{n-i} \quad \text{for } n \geq 0$$
- Euler Characteristic:
planar graph: $V - E + F = 1$
convex polyhedron: $V - E + F = 2$
 V, E, F, C : number of vertices, edges, faces(regions), and components
- Kirchhoff's theorem :
 $A_{ii} = \deg(i), A_{ij} = (i, j) \in E ? -1 : 0$, Deleting any one row, one column, and cal the $\det(A)$
- Polya' theorem (c is number of color, m is the number of cycle size):
$$\left(\sum_{i=1}^m c^{\gcd(i, m)} \right) / m$$
- Burnside lemma:
$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |X^g|$$
- 錯排公式: (n 個人中, 每個人皆不再原來位置的組合數):
$$dp[0] = 1; dp[1] = 0;$$

$$dp[i] = (i-1) * (dp[i-1] + dp[i-2]);$$
- Bell 數 (有 n 個人, 把他們拆組的方法總數) :
$$B_0 = 1$$

$$B_n = \sum_{k=0}^n s(n, k) \quad (\text{second - stirling})$$

$$B_{n+1} = \sum_{k=0}^n \binom{n}{k} B_k$$
- Wilson's theorem :
$$(p-1)! \equiv -1 \pmod{p}$$
- Fermat's little theorem :
$$a^p \equiv a \pmod{p}$$
- Euler's totient function:
$$A^{B^C} \bmod p = \text{pow}(A, \text{pow}(B, C, p-1)) \bmod p$$

- 歐拉函數降冪公式:
$$A^B \bmod C = A^{B \bmod \phi(C) + \phi(C)} \bmod C$$
- 6 的倍數:
$$(a-1)^3 + (a+1)^3 + (-a)^3 + (-a)^3 = 6a$$
- Standard young tableau (標準楊表):
 $\lambda = (\lambda_1 \geq \dots \geq \lambda_k), \sum \lambda_i = n$ denoted by $\lambda \vdash n$
 $\lambda \vdash n$ 意思為 λ 整數拆分 n eg. $n = 10, \lambda = (6, 4)$ 此拆分可表示一種楊表形狀。
楊表: 第 1 列 λ_1 行 $\dots$ 第 k 列 λ_k 行的方格圖。
標準楊表: 每列從左到右遞增, 每行從上到下遞增。
Let T 為某一 Permutation 跑 RSK 後的標準楊表, 則此 Permutation 的 LDS、LIS 長度分別為 T 的列、行數。
- RSK Correspondence:
A permutation is bijective to (P, Q) 一對標準楊表
 P : Permutation 跑 RSK 算法的結果, 可為半標準楊表。
 Q : 可用來還原 Permutation (像排列矩陣)。
- Hook length formula (形狀為 λ 的標準楊表個數):
$$f^\lambda = \frac{n!}{\prod h_\lambda(i, j)}$$

 $h_\lambda(i, j) = \text{number of pair } (x, y) \text{ where } (x = i \vee y = j) \wedge (x, y) \geq (i, j)$
且 (x, y) 落在形狀為 λ 的表上。
Recursion:
(i) $f^{(0, \dots, 0)} = 1$
(ii) $f^{(\lambda_1, \dots, \lambda_m)} = \sum_{k=1}^m f^{(\lambda_1, \dots, \lambda_{k-1}, \lambda_k-1, \lambda_{k+1}, \dots, \lambda_m)}$

9.2 Prime

不知道要做啥但放一下

Prime	Root	Prime	Root
7681	17	167772161	3
12289	11	104857601	3
40961	3	985661441	3
65537	3	998244353	3
786433	10	1107296257	10
5767169	3	2013265921	31
7340033	3	2810183681	11
23068673	3	2885681153	3
469762049	3	605028353	3
924844033	5	0	0



為什麼要演奏春日影!

